INSTALLATION

(1) Position the rear washer nozzle and rubber gasket to the outside of the liftglass with the orifice of the nozzle aimed at the nozzle alignment arrowhead visible on the outside of the glass.

(2) From the inside of the liftglass, install and tighten the plastic nut that secures the threaded nipple on the back of the rear washer nozzle to the inside of the liftglass (Fig. 9). Tighten the nut to 1 N·m (8 in. lbs.).

(3) Reconnect the rear washer supply hose to the barbed nipple of the rear washer nozzle.

(4) Reinstall the trim cover onto the rear wiper motor on the inside of the liftglass. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER MOTOR TRIM COVER - INSTALLATION).

REAR WASHER PUMP MOTOR

DESCRIPTION

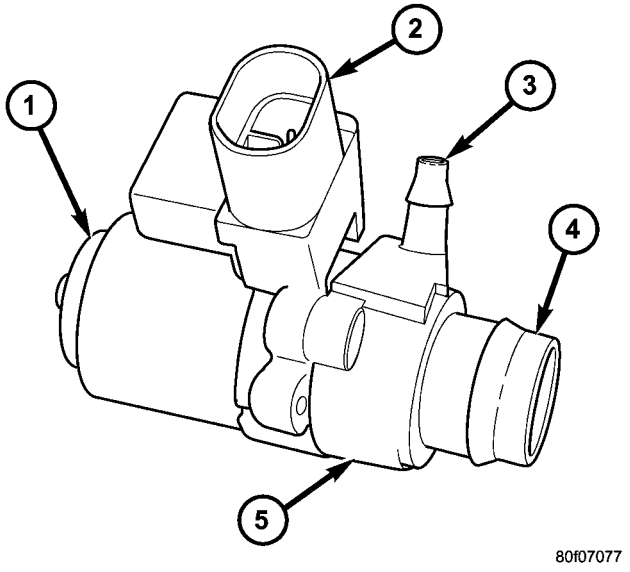

Fig. 10 Washer Pump/Motor

1 - MOTOR
2 - CONNECTOR RECEPTACLE
3 - OUTLET NIPPLE
4 - INLET NIPPLE
5 - PUMP

The rear washer pump/motor unit is located on the inboard side just behind and above the front washer pump/motor on the washer reservoir (Fig. 10). The washer reservoir is located on the top of the left front fender wheel house in the engine compartment. A small permanently lubricated and sealed electric motor is coupled to the rotor-type washer pump. A seal flange with a barbed inlet nipple on the pump housing passes through a rubber grommet seal installed in a dedicated mounting hole in the sump area near the bottom of the washer reservoir. The rear washer pump/motor unit is always mounted in the rearward-most upper mounting hole of the reservoir. An integral filter screen is located within the pump inlet nipple. When the pump is installed in the reservoir a barbed outlet nipple on the pump housing connects the unit to the washer system through the front washer supply hose.

The washer pump/motor unit is retained on the reservoir by the interference fit between the barbed pump inlet nipple and the grommet seal, which is a light press fit. An integral connector receptacle between the motor housing and the pump housing connects the unit to the vehicle electrical system. A small capacitor internal to the connector receptacle insulator is connected in parallel with the motor leads to control ElectroMagnetic Interference (EMI) created by washer motor operation. The rear washer pump/motor unit cannot be repaired. If faulty or damaged, the entire washer pump/motor unit must be replaced.

OPERATION

The rear washer pump/motor unit features a small Direct Current (DC) electric motor. The motor is connected to the vehicle electrical system through a single take out and two-cavity connector of the body wire harness. The motor is grounded at all times through a take out of the body wire harness with a single eyelet terminal connector that is secured under a ground screw to the left cowl side inner panel below the instrument panel in the passenger compartment. The motor receives battery current through the closed contacts of the momentary rear washer switch circuitry within the rear wiper and washer switch unit only when the bottom of the switch rocker is pushed towards the instrument panel.

Washer fluid is gravity-fed from the washer reservoir to the inlet side of the washer pump. When the pump motor is energized, the motor spins the rotor within the rotor-type washer pump. The spinning pump rotor pressurizes the washer fluid and forces it through the pump outlet nipple, the rear washer plumbing, and the rear washer nozzle onto the liftglass. The rear washer pump/motor unit can be diagnosed using conventional diagnostic tools and methods.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Disconnect the washer supply hose from the barbed outlet nipple of the rear washer pump/motor and allow the washer fluid from the washer reservoir to drain into a clean container for reuse (Fig. 11).